CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of Master of Technology (M. Tech)
University Examination May 2011
CE711 Information and Network Security

Date: 17.05.11 **Time:10.00 a.m to 1.00 p.m** **Maximum Marks:70**

Instructions:

1. *Attempt all questions.*
2. *Answers both sections in separate answer books.*
3. *Figure to the right indicates full marks.*
4. *Assume suitable data, if necessary and mention it.*

SECTION I

Q-1

(a) Differentiate between passive and active security attacks. Name some passive attacks and active attacks. **3**

(b) Show the matrix representation of transposition-cipher encryption key with the key (3, 2, 6, 1, 5, 4). Find the matrix representation of the decryption key. **4**

(c) Encrypt the message "meet me at the usual place at ten rather than eight oclock" using the Hill cipher with the key $\begin{bmatrix} 4 & 2 \\ 1 & 1 \end{bmatrix}$. Show your calculations and the result. Apply same algorithm to find plain text from the encrypted text. **4**

Q-2

(a) To be resistant to exhaustive-search attack, a modern block cipher needs designed as a substitution cipher. State true or false. Justify your answer with example. **4**

(b) Create a Linear Feedback Shift Register (LFSR) with 10 cells in which having $b_{10}=b_9+b_5+b_4+b_2+b_0$. Find out maximum period length of an LFSR. Show encryption process using 100011101110111 plain text. **4**

(c) How many exclusive operations are used in the DES cipher? Why does the DES function need an expansion permutation? Why does the round key generator need a parity drop permutation? **4**

OR

Q-2

(a) Explain the different types of attacks on block cipher. Describe the difference between stream ciphers and block ciphers. **4**

(b) What is triple DES? What is triple DES with two keys? What is triple DES with three keys? **4**

(c) Explain the types of transformation that Advanced Encryption Standard (AES) uses in each round. **4**

Q-3

(a) List the mode of operations used in symmetric cryptosystem. Write the advantages and disadvantages of various modes of operations. **5**

(b) Explain the four objectives of blowfish algorithm. Describe how sub key generation process works in it. **7**

SECTION II

Q-4

(a) Why symmetric key distribution is crucial? What is session key? Explain working of ticket granting server and authentication server in Kerberos. **7**

(b) Explain the difference between symmetric key cipher and asymmetric key cipher algorithms. List the name of Feistel ciphers which are symmetric key ciphers. **4**

Q-5

(a) What is the meaning of signed document in cryptography? Why asymmetric key is generally used for signing the document? **3**

(b) Describe the various ways of public-key distribution. Explain the format of a X.509 version 3 certificate. **6**

(c) Given a super increasing tuple b = [7, 11, 23, 43, 87, 173, 357], r=41 and modulus n=1001, encrypt and decrypt the word "abc" using the knapsack cryptosystem. Use [7 6 5 1 2 3 4] as a permutation table. **3**

OR

(c) In RSA: **3**

a. Given n=221 and e=5 find d.
b. Given n=3937 and e=17 find d.
c. Given p=19, q=23 and e=3, find n, Ø(n) and d.

Q-6 **Write a short note on any two:** **12**

(a) Pretty Good Privacy (PGP).
(b) S/MIME
(c) SSL
(d) IPSec